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(57) Abstract :

ABSTRACT GENERATIVE AI CHATBOT FOR STUDENT EMOTIONAL WELL-BEING WITH BURNOUT RISK DETECTION AND ADAPTIVE COPING ENGINE The present invention discloses a generative AI-based chatbot system (100) for student emotional well-being comprising a user interface module (110), an NLP processing unit (120) for text preprocessing, an emotion detection module (130) employing BERT-based transformer models for classifying user input into emotional categories of stress, anxiety, sadness, fatigue, and motivation with sentiment polarity and intensity determination, a burnout risk detection engine (140) computing a composite weighted burnout score using a multi-parameter formula incorporating emotional intensity, frequency of negative input, productivity, and engagement to categorize users into low, moderate, or high risk levels, a generative AI response module (150) employing GPT-based transformer models for generating context-aware empathetic responses, an adaptive coping strategy engine (160) providing personalized evidence-based recommendations with continuous learning from feedback, an emotion-productivity analytics module (170) correlating emotional patterns with academic productivity, a database module (180) for data persistence, and a backend server module (190) for inter-module communication. [to be published with fig.1]

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